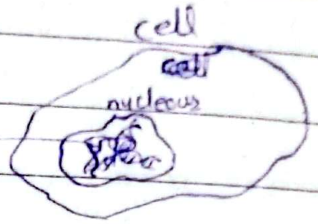
Date Day

BASICS OF BIOLOGY



① Cell

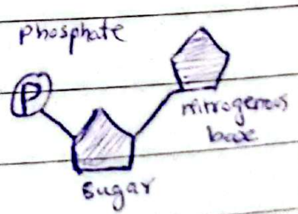
- The basic unit of life
- Each Cell has a structure, performs function contains DNA.
- Each cell has a different kind of function, How does a cell know which func to perform?
  - DNA tells it. (like an instuction manual)

② DNA (Deoxyribonucleic Acid)

- DNA is found inside the nucleous of a cell
  - Brain of a cell
- It is a long molecule that stores biological instruction

• Structure of DNA

- DNA is made up of nucleotides.
  - Nucleotides are building blocks or "letters" of DNA
  - Each nucleotide has 3 parts



① Sugar

② Phosphate Group

③ Nitrogenous Base - 4 types

i- Adenine (A) iii- Cytosine (C)

ii- Thymine (T) iv- Guanine (G)

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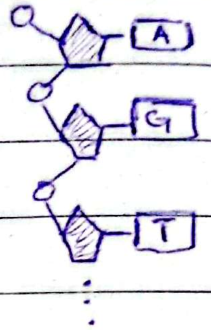
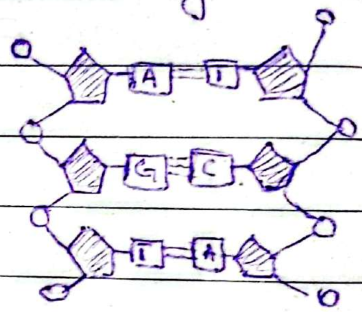
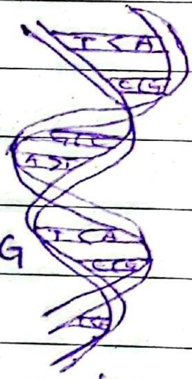
- A single DNA Strand
  - ↳ Many nucleotides join together through phosphate - sugar bonds to form a single strand of DNA



- Double Helix - Classic DNA Shape
  - ↳ DNA is double-stranded, meaning there are two single strands twisted around each other





RULE

→ A only pairs with T

→ C only pairs with G

→ This is called a base pair

Strand 1: A T G C ⟶ DNA sequence

Strand 2: T A C G

↳ If we have one strand then we can easily know the other

→ Each human cell contains 46 strands of DNA

→ DNA is extremely long — millions of letters

→ It stays inside the nucleus of a cell

→ For sequence lengths we use

Kb → kilo - bases , Gb → Giga bases

Mb → Mega - bases

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▷ Amino Acid
↳ Amino acid are small chemicals in the body
↳ They are called "building blocks of life"
↳ They are about 20 diff & types
↳ Each has unique shape
↳ They connect together like legos
↳ Amino acid forms → protiens

▷ Protiens
→ protien is also represented by String of letter. Each letter is an amino acid which comes thru codon.
↳ Protiens are made of chains of amino acid
↳ There are millions of diff protiens
↳ Protient must be fold into correct shape to work properly
↳ If shape is wrong → protien will not function

Amino Acid → Protien → Cells → Tissues → Organs

▷ How Does DNA Works?
↳ DNA has many functions
↳ One of its main function is to tell amino acids how to line up to make protiens
↳ If the right protiens are made at the right time and right place
↳ Cells work properly
↳ Organs form ч.

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- DNA → RNA → Protien
  - DNA cannot leave nucleous
  - But it has to instruct amino acid for correct formation of protien
  - Amino acid is in Cytoplasm
    - Area outside nucleus
  - Special chemicals make a partial copy of DNA called RNA (Ribonucleic Acid)
  - RNA is shorter than DNA
    - It is single stranded
    - RNA leaves the nucleus thru tiny pores
    - RNA uses U (Uracil) instead of T.
  - RNA goes to a structure called Ribosomes
  - Ribosomes are protien building machine
  - Ribosomes reads RNA 3 letters at a time
    - Every 3 letter = One amino acid instruction
      - codons (mentioned later in notes)

- What is a Gene.
  - A gene is a special segment of DNA
  - It is a specific sequence of A, T, C, & G.
  - A gene is not the entire DNA strand
  - It is just a small section of DNA
  - Genes makes protein
  - One DNA strand contains thousands of genes
  - Humans have about 20.000 genes in total

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↳ Gene size varies
    ↳ Some are only 300 letters long
    ↳ " " over 1 million " ".
↳ The length and sequence determine
    ↳ Protien size , Protien Shape , Protien functio

↳ Different Organisms have diff genes diff protien recipes
But,
All life uses the same DNA language (A, T C, G)

▷ **Chromosomes**
↳ DNA is extremely long
↳ To fit inside the cell, it is tightly packed into structures called chromosomes
↳ ~~DNA~~ Misconception:
DNA is not stored inside a chromsome
Instead : The Chromosome ITSELF is the DNA, tightly wrapped
↳ Humans have 23 pairs of Chromosomes
↳ 46 total in each cell
↳ For each pair 1 Chromosome comes from mother & another from father.
↳ Together these make up genome

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→ Genome is a complete set of DNA inside a cell of organism.

→ Almost every cell in your body already contains full copy of your genome

→ But different cells turn diff genes ON or OFF

→ Example:

↳ Eye cells use eye-related genes

↳ Liver " " liver " "

↳ and so on.

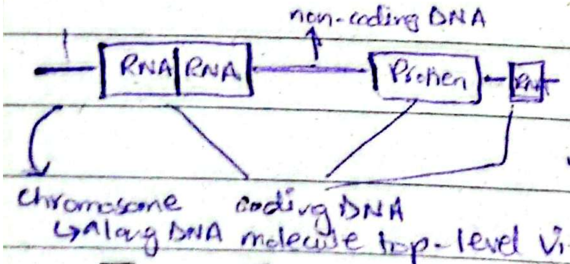
→ This is called gene regulation.

↳ Genome comprises the coding & non-coding genetic sequence

CDNA (coding DNA)

↳ Coding Sequence — Part of genome that encodes protien

↳ Non-coding " — Part of genome that does not encode protien but still has some function which are not known yet.



→ **Encoding**

↳ The sequence of DNA bases determines the sequence of amino acids in a protien.

→ A protien string AEFFQP (each alphabet is called residue)

→ A codon tells amino acid how to line up to form protien

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↳ Every 3 DNA bases = 1 amino acids

↳ Example:

- A DNA bases ATG codes for amino acid Methionine
- This 3-base unit is called a **codon**

DNA	RNA
→ Long and usually double stranded	→ Short & often single stranded
→ Trapped inside the nucleus	→ Active all throughout the cell
→ Relatively stable	→ ~~Relatively~~ High Fragile
→ Guarded & Repaired when damaged	→ Digested & rebuilt when damaged
→ Used for information storage	→ Used for active functions such as protien coding
→ Base pair has ~~the~~ T	→ It has U (Uracil)

→ As 3-base unit forms → 1 amino acid so it mean we should have

$\underset{A/C/G/T}{4} \times \underset{A/C/G/T}{4} \times \underset{A/C/G/T}{4} = 64$ combination of codons so 64 Amino acids But in reality there are only $\boxed{20}$ amino acids. (because we have some redundancies & special cases)

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Q How to determine start & end of a gene in a DNA sequence?

↳ The cell knows exactly from where a gene starts & end.

↳ There are specific codons that determine the start of Gene.

Start : ATG (DNA) / AUG (RNA)

↳ codes for Methionine

End : TAA, TAG, TGA

↳ They donot code for any protien.

↳ When ribosome sees these codon, protien synthesis stops